

AutismCarebot: Emotion-First, Source-Aware Conversational Support for Autistic Users

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Abstract

Autistic people often seek support for stress regulation, communication challenges, and navigating services, yet general-purpose chatbots may miss neurodiversity-affirming language and can provide ungrounded advice. We present AutismCarebot, a web-based conversational agent that combines (1) an emotion-first response structure (validation, normalization, and a clear next step) with (2) retrieval-augmented generation (RAG) to provide concise guidance grounded in curated autism resources with in-text citations. We designed the interaction to reduce cognitive load through quick prompts, progressive disclosure of detail, and a built-in “Take a Break” module offering guided breathing and grounding. We evaluated AutismCarebot through an automatic comparison with an unmodified large language model baseline and a survey study (n=18) including autistic adults, caregivers, educators, and students. Results indicate AutismCarebot produced more empathetic and better-grounded responses than the baseline, with no unsafe outputs observed in the audit. The survey feedback highlighted AutismCarebot’s soothing tone, clarity, and practical advice, while emphasizing the necessity to control the level of links and provide multimodal support.

CCS Concepts

• **Human-centered computing**; • **Empirical studies in HCI**;
• **Applied computing**; • **Health care information systems**; •
Computing methodologies; • **Natural language generation**;

Keywords

autism, neurodiversity, conversational agents, digital mental health, retrieval-augmented generation, source transparency, empathetic interaction

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1 Introduction

Autistic people may seek support for managing stress, navigating social demands, and understanding services, often while contending with sensory overload and a preference for predictability and routine [1]. Digital tools can provide private, on-demand assistance, but generic conversational agents are not designed for neurodiversity-affirming communication and may produce overconfident, ungrounded guidance-risks that are particularly concerning in mental health contexts [2]. Existing mental health chatbots are typically aimed at general audiences and rarely offer autism-specific interaction support.

This paper presents AutismCarebot, a conversational agent designed to provide emotional and cognitive support in the situations that autistic individuals often encounter, such as feeling overwhelmed, experiencing shutdown, exhibiting communication breakdowns, and needing service navigation, while prioritizing trust and user control. AutismCarebot combines an emotion-first response structure with retrieval-augmented generation (RAG) to ground advice in curated resources and surface citations alongside suggestions. The interface includes quick prompts, progressive disclosure of detail, and a built-in “Take a Break” module to support self-regulation when users feel overwhelmed.

Our contributions are: (1) an interaction design that operationalizes emotional acknowledgement, normalization and structured guidance for autistic users (Figure 1); (2) a source-aware RAG pipeline, which returns succinct replies with citations to minimize the risks of misinformation; and (3) an evaluation of automatic metrics, a manual audit, and preliminary user survey (n=18) of autistic people and supporters.

2 Related Work

Health and mental health conversational agents are designed to support a person, but systematic reviews of personalization in health conversational agents highlight diverse implementations and limited evaluation of personalization’s impact [3]. Simultaneously, retrieval-augmented generation (RAG) emphasizes factual grounding and can render model outputs more traceable [4], whereas transparency is empathy prioritized in HCI research on conversational health agents [19]. Mental health professionals value the scalability of chatbots, but stress that clear boundaries, safety nets, and emergency procedures are also necessary [20]. Sharma et al. show that breaking down mental health tasks into smaller, more manageable steps and giving the user control over the interaction can be incredibly effective in a chat-based therapy setting [21].

Regarding situations which individuals with neurodiversity and anxiety disorders often encounter, recent works in CHI and

PACMHCI show that LLM-based support has a lot to offer but also has its limits. For example, Carik et al. found that neurodivergent people are using LLMs for planning, rewriting, and social scripting, but users worry that this might lead to overreliance on the technology, and unpleasant results [18]. Users' trust of generative AI for help with social anxiety depends on its perceived competence, transparency and how well it gets along with our personal values [19].

Conversational agents designed for autistic users emphasize the critical role of empathy, tone, and role clarity [2]. However, research cautions against the possibility of generic or too clinical responses failing to address the needs of autistic users [2] as some of these agents can repeat deficit-based framings and may provide advice in the absence of enough background knowledge, which can be damaging for trust and security [2]. There is still limited guidance on how to combine (1) emotion-first support, (2) source-grounded suggestions, and (3) overload-sensitive presentation for autistic users—especially when citations and links themselves can become overwhelming.

These challenges and the preference among autistic users for routine and predictability are interrelated [1]. Consequently, the rigidity of interaction structure and the clarity of the information will be important, above the capability of the model. Research on mental health chatbots for autistic users builds upon research investigating how autistic people use LLMs in one-to-one conversations. Autistic people use ChatGPT-like tools in their daily routines for tasks such as interpersonal communication assistance (e.g., rewriting emails to adjust tone or social scripting), learning and information synthesis, professional development, and emotional well-being support [18, 22]. Predictability and user control contribute to usable AI systems for autistic users [8, 9]. Work-related LLMs could be used for communication support in the workplace, highlighting the need for clarity, agency, and manageable interactions [23]. Earlier research has explored how autistic people use general-purpose language models in their daily lives, and how AI tools can help with communication and accessibility. AutismCarebot builds on that foundation by bringing several ideas together in one system: an emotion-first response style, guidance grounded in trusted sources with carefully limited citations, and an interface designed to reduce cognitive overload.

3 AutismCarebot System

3.1 Design Goals

The user interface (UI) of AutismCarebot was designed to provide an inclusive and calming environment for autistic people. We focused on clarity, simplicity, and consistency to lessen cognitive overwhelm that can be caused by complex images and abrupt changes. Based on neurodiversity design guidelines [24] and prior autism-centered HCI work on designing AI for autistic adults [8], we articulated and followed into four design goals:

- **DG1:** Emotional safety first. Normalization and validation should be done first and then strategies should be provided.
- **DG2:** User control and predictability. Offer choices (e.g., **quick prompts and optional resources**) and keep responses structured (Figure 1).

- **DG3:** Trustworthy information with traceability. Ground guidance in curated sources and attach citations in a structured manner.
- **DG4:** Overload-sensitive presentation. Use concise language, quick prompts, and optional self-regulation tools.

3.2 Emotion-First Interaction Flow

AutismCarebot follows a consistent response pattern: (1) acknowledge the emotional situation of a user, (2) normalizing the experience, (3) offer **optional guidance** with a clear, structured next step, and (4) proposing a single definite next step and the optional ones (Figure 2). The design follows an emotion-first pattern (validation, normalization, then a concrete next step), informed by prior work on empathy in dialogue systems [10], and offers predictability to the users, who might be stressed with unstructured responses [1].

To implement this pattern, we fine-tune Meta's Llama 3.2 3B Instruct using QLoRA [6, 7]. When the user's message contains indicators of acute distress, a safety layer prioritizes grounding exercises and crisis resources over informational content.

3.3 Source-Aware Suggestions with Citations

Users and caregivers often want to know where advice comes from, especially when a chatbot makes claims about autism, services, or coping strategies. AutismCarebot therefore pairs generation with retrieval: a curated corpus of public autism-related resources selected from national health agencies, professional associations, and autism organizations is chunked and embedded; at run time, the system retrieves the most relevant passages and conditions the model on them [4]. As shown in Figure 2, the response includes an *Additional Resources* section, with numbered, clickable citations linked to the underlying sources, along with a short *References* list of URLs.

To reduce overload, AutismCarebot limits citations in the main response (typically 1-3), and places additional links in the 'Additional Resources' section to support traceability while limiting overload.

3.4 Overload-Sensitive Interface Features

The web interface supports multiple interaction styles. Users can start with quick prompts (e.g., "I feel overwhelmed," "Help me explain my needs to someone"), use speech-to-text, or request shorter or more detailed answers. A dedicated Take a Break module offers short, guided self-regulation, including paced breathing exercises, grounding exercises, and so on, which the user can use immediately in case they are stuck or overstimulated.

4 Evaluation

We conducted a mixed-methods evaluation with the goal to (a) compare the quality of responses and empathy to an unmodified baseline model, (b) factual grounding and safety by having a manual audit, and (c) the initial perceptions of usefulness and usability by users.

4.1 Automatic and Audit-Based Evaluation

The experiment was an attempt to compare AutismCarebot to the original Llama 3.2 3B Instruct baseline [7] with a common prompt

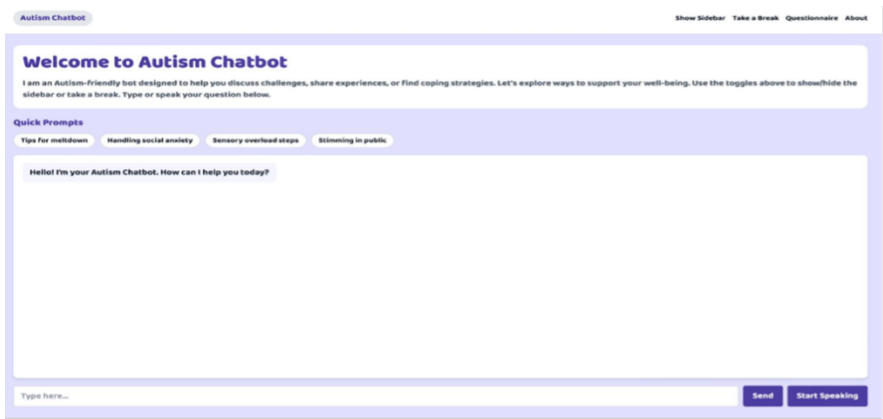


Figure 1: AutismCarebot prototype UI. Calm interface with quick prompts and optional access to additional content (e.g., resources) to reduce clutter and cognitive load.

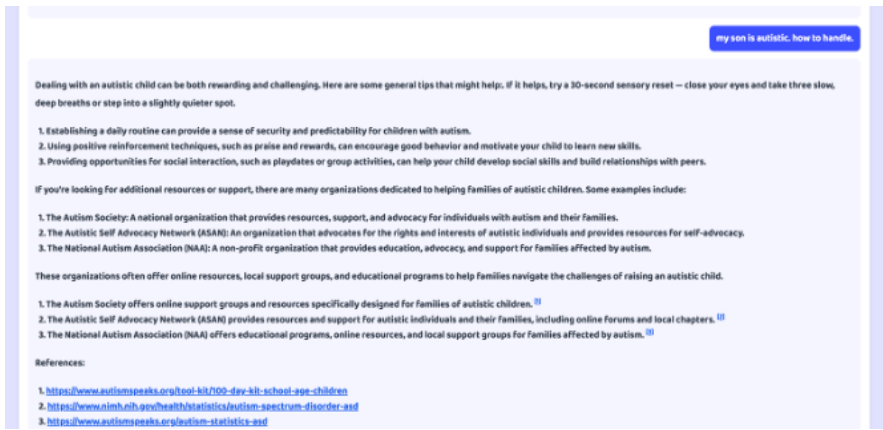


Figure 2: Example source-aware response. Example caregiver query and response (“My son is autistic-how do I handle it?”) illustrating emotion-first support and traceability via numbered, clickable in-app citations and linked reference URLs.

Table 1: Summary of automatic metrics and audit outcomes comparing AutismCarebot with an unmodified baseline model. Automatic metrics are computed against empathy-optimized target responses; citations are removed prior to scoring.

Metric	Baseline	AutismCarebot
BLEU-4 (lexical match)	0.022	0.433
ROUGE-L (F1)	0.113	0.576
BERTScore (F1)	0.702	0.774
Empathy score (0–1)	0.450	0.986
Latency (mean, s)	24.5	20.0
Unsafe response rate (audit)	-	0/107 flagged unsafe (95% upper bound $\approx$ 2.8%)

template and decoding configuration. Across 57 prompts with empathy-aligned target responses, authored by the research team and informed by autistic-user feedback and prior autism-centered HCI work. We report BLEU-4/ROUGE-L [11, 12] for lexical overlap, BERTScore [13] for semantic similarity, and an author-defined empathy rubric capturing validation, reflective listening, normalization, and one concrete next step [5]. Citations were removed and

only the first two sentences were scored. These target responses served only as evaluation anchors for automatic text-overlap metrics and were not treated as ground truth.

In addition to automatic metrics (Table 1), we manually audited 107 responses for unsafe content. No responses were flagged unsafe; using the “rule of three,” this corresponds to a 95% upper bound of

approximately 2.8% given $N=107$ [15]. Additionally, we used well-established test suites for LLM risks to spot-check safety behavior [16, 17].

4.2 Usability Study

We conducted a usability study ($n=18$) to evaluate the perceived empathy, helpfulness, and usability. Participants were recruited using snowball sampling and the study had IRB approval. The analyzed sample included 6 autistic adults, 4 caregivers, 2 educators, 4 students, and 2 other users. The online study comprised using AutismCareBot and filling out a form of open-ended questions about what they found most and least valuable and the **10-item System Usability Scale (SUS)** [14] (5-point Likert), along with open-ended questions.

The results of the study show strength in usability and fostering a calming environment. Participants reported positive usability and a calming experience. On the 10-item System Usability Scale (SUS) [14], responses to positive items (e.g., easy to use, learn quickly, confidence) skewed toward Agree/Strongly Agree, while reverse-worded items (e.g., inconsistency/complexity) skewed toward Disagree/Neutral. Participants also reported that the advice felt actionable and that they felt more reassured or informed after the chat. Users self-reported that they valued AutismCareBot's calming voice and practical tips. The Take a Break feature was often reported as useful in anxiety reduction. Participants requested accessibility supports (text-to-speech, increased visuals, etc.), and some participants remarked that the chatbot's answers, which are full of links, may be overwhelming, which once again supports the argument of citing carefully.

5 Discussion

According to our findings, a combination of source grounding and an emotion-first interaction technique can enhance the quality of both felt support and quantifiable responses. We emphasize three design considerations for conversational agents that promote neurodiversity:

Establish validation as the default setting. Begin with an emotional assessment (validation and normalization) prior to transitioning into guidance.

Ground suggestions but throttle citations. Provide a small number of sources by default and keep additional references in a dedicated section.

Design for pacing and predictability. Use consistent structure, and stepwise self-regulation supports to reduce overload [1].

5.1 Conclusion

The concept of AI-based conversational systems is becoming widely accepted as an efficient method of addressing mental health provision gaps, in particular where individualized, scalable and affordable care is demanded. In this paper, we created AutismCarebot, an empathy-enhanced and retrieval-augmented conversational chatbot adapted to the unique communication, emotional, and sensory demands of patients with Autism Spectrum Disorder (ASD). With the Retrieval-Augmented Generation (RAG) and a well-tuned LLaMA-3.2 model based on QLoRA, the system manages to balance empathetic dialogue generation and information accuracy based

on trusted sources of knowledge related to autism. The experimental findings and user experiment suggest that AutismCarebot showed improvements relative to a baseline mental health chatbot in dialogue quality, semantic relevance, and emotional alignment, as shown by substantial improvements across automatic metrics and user-reported empathy. Additionally, the qualitative feedback and usability testing has shown that the autism-friendly design of the interface, predictive prompts, and supportive conversational strategies are effective at mitigating the information overload and anxiety, which creates a feeling of emotional safety and trust.

5.2 Limitations and Future Work

This work is an early step. As the pilot user study was small and short-term, future work should consider longitudinal deployments to understand sustained use, habituation, and potential harms. Additionally, the curated corpus reflects the authors' sourcing choices and may omit culturally specific or community-led materials. It is important for the future works to expand sources, offer user-configurable source preferences, add multimodal supports (e.g., text-to-speech and visual summaries), and further evaluate crisis-handling response in realistic stress conditions. To ensure that the research is globally applicable, the RAG knowledge base might be expanded with region-specific and multicultural autism resources. Finally, utilizing explainable AI methods and clinician-in-the-loop assessment frameworks would keep the responses transparent and trustworthy, while enhancing clinical relevance, leading to wider implementation of specialized conversational AI systems in neurodiversity-affirming mental health care.

6 Ethics and Disclosure

AutismCarebot is not a medical device and functions strictly as an informational tool. It does not provide diagnosis or emergency services. We have ensured that the system includes boundary-setting language, encourages consultation with qualified professionals when appropriate, and prioritizes crisis resources when users indicate self-harm or imminent danger. Regarding data privacy and ethical development, all human-participant procedures were reviewed in the authors' research environment, and responses were collected and analyzed in de-identified form.

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